

Background and 511 keV line sensitivity of a balloon-borne TES Laue-lens telescope^{*}

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Abstract

The Galactic 511 keV annihilation line is a direct tracer of low-energy positrons, but its compact-source content and narrow-line kinematics remain difficult to measure because MeV instruments are strongly background dominated. We present a full-chain Monte Carlo and rate-folding study for a balloon-borne pointed 511 keV telescope that combines a Ge(111) Laue lens with a transition-edge-sensor (TES) microcalorimeter focal plane. The model couples the focused optical phase space to an engineering detector/cryostat geometry, prompt atmospheric backgrounds, activation buildup, exact-position delayed-source transport, active-shield vetoes, Compton/field-of-view consistency cuts, a narrow line window, and mission-time rate variation. The current v3p5 full-stat exact-position chain uses a 10 m focal-length Laue configuration with $A_{\text{eff}}(511 \text{ keV}) = 20.08476 \text{ cm}^2$ and a 510.58–511.42 keV analysis window. At a reference point-source flux of $10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$, the selected line-window background and signal rates are 0.0624651 s^{-1} and $0.00118117 \text{ s}^{-1}$, respectively. The mission-time fold gives $Z_{20\text{d}} = 6.15522$, a 3σ detection time of 4.73758 d, a 20-d 3σ flux threshold of $4.87391 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$, and a 1 Ms threshold of $6.32564 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$. A transport-backed exact-position support-size and seed sweep changes the total line-window background by only 1.119% and $Z_{20\text{d}}$ by 0.551%. We also report spatial-analysis and atmospheric-line-of-sight sidecars, while keeping the hard-window exact-position chain as the primary rate authority. The results indicate that focused TES spectroscopy can reach scientifically useful narrow-line sensitivity from a balloon platform, provided activation and prompt atmospheric backgrounds are modeled in the same detector-coupled selection space as the

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signal.

Keywords: Gamma-ray instrumentation, 511 keV annihilation line, transition-edge sensors, Laue lens, balloon-borne telescope, Monte Carlo background simulation, activation background

1. Introduction

The 511 keV gamma-ray line from electron–positron annihilation is one of the cleanest spectroscopic signatures of antimatter in the Galaxy. After positrons slow down in the interstellar medium, they annihilate either directly with electrons, producing two photons near 511 keV, or through positronium formation, producing the 511 keV line together with a three-photon continuum below the line. This radiative complex probes positron production, propagation, thermalization, and annihilation conditions [1]. The line was first reported from the Galactic-center direction by balloon-borne experiments and later identified unambiguously with high-resolution Ge detectors as positron-annihilation radiation [1]. The observed Galactic flux implies a positron annihilation rate of order $10^{43} \text{ e}^+ \text{ s}^{-1}$, making the origin of these positrons a long-standing problem in MeV gamma-ray astronomy [1, 4].

The candidate source classes span standard and non-standard physics. Conventional astrophysical candidates include β^+ decay of radioisotopes synthesized in massive stars, novae, core-collapse supernovae, and thermonuclear supernovae; pair creation in compact objects such as X-ray binaries, microquasars, pulsars, and the Galactic-center black hole; and cosmic-ray interactions [1, 2]. More speculative explanations include light dark matter decay or annihilation and hidden-sector scenarios [1, 3]. The difficulty is that the observed annihilation morphology is shaped not only by where positrons are produced, but also by how far they propagate before thermalization and annihilation. Spectroscopy constrains the annihilation medium through the line centroid, line width, and positronium fraction, while continuum limits above 511 keV constrain the injection energy [1, 2]. The most discriminating observables for source identification—compact-source content, small-scale structure, velocity shifts, and spatially resolved line broadening—remain limited by instrumental sensitivity, angular resolution, and background systematics.

The current observational picture has been shaped by OSSE/CGRO, INTEGRAL/SPI, and COSI. INTEGRAL/SPI established a bright bulge-like

32 component and fainter disk emission, with the detailed bulge/disk decom-
 33 position depending on the adopted sky model and instrumental-background
 34 treatment [2, 3]. Recent long-exposure SPI analyses recover the central,
 35 broad-bulge, and disk structures and suggest marginal additional structures
 36 associated with massive-star regions [3]. COSI provided an important in-
 37 dependent measurement from a 46-d super-pressure-balloon flight, detecting
 38 the Galactic 511 keV signal with Compton imaging and finding that the emis-
 39 sion is broader than a single point source [4, 5]. These results demonstrate
 40 both the scientific value and the practical difficulty of MeV line astronomy:
 41 the celestial line is weak, while atmospheric and instrumental backgrounds
 42 are strong and time dependent.

43 Wide-field Compton telescopes are essential for mapping the large-scale
 44 511 keV sky. A complementary strategy is to use a pointed telescope that
 45 concentrates 511 keV photons onto a small, high-resolution focal plane. A
 46 pointed system is especially useful for testing whether a fraction of the anni-
 47 hilation luminosity is produced by unresolved point sources, a bright central
 48 core, or transient pair-plasma episodes in compact objects. This motivation
 49 is reinforced by annihilation signatures reported during the 2015 outburst of
 50 V404 Cygni, which indicate that accreting stellar-mass black holes can pro-
 51 duce pair plasma at astrophysically relevant rates [6]. Such a payload does
 52 not replace wide-field Compton missions; it addresses a different observable
 53 in a narrower field of view.

54 The 511-CAM concept follows this route by combining gamma-ray con-
 55 centrating optics with stacked transition-edge-sensor microcalorimeter arrays
 56 [7]. It is attractive because it joins photon concentration with sub-keV calori-
 57 metric energy resolution, two capabilities rarely available together at MeV
 58 energies. A projected 511 keV detector-assembly resolution of a few 10^2 eV
 59 FWHM corresponds to a velocity scale of a few 10^2 km s $^{-1}$ and could enable
 60 velocity-shift or line-broadening measurements in favorable cases [7]. Event
 61 topology also matters. Single-pixel photoelectric events provide the clean-
 62 est spectroscopy, while multi-pixel events can be checked against Compton
 63 kinematics and rejected if inconsistent with photons arriving through the
 64 focusing optics.

65 The central technical challenge is background. Balloon-borne gamma-ray
 66 spectrometers operate in a strong radiation environment and are necessarily
 67 background dominated. Classical studies of balloon-borne Ge spectrometers
 68 identify atmospheric-neutron scattering, atmospheric and cosmic gamma-ray
 69 aperture flux, beta decays of activation products, and shield leakage as major

background components; they also emphasize minimizing passive material near the detector and lowering effective shield thresholds where possible [8]. For a 511 keV line payload, the problem is sharper because the science signal lies at the same energy as strong atmospheric and instrumental annihilation features. A credible sensitivity estimate therefore requires a detector-coupled model of the optics, entrance windows, cryostat, passive material, active shield, prompt atmospheric components, and delayed activation.

Detailed Monte Carlo mass modeling is the standard route to such estimates. The INTEGRAL/SPI response production required a high-fidelity mass model, ground- and in-flight-calibration comparisons, and a practical separation between imaging response and redistribution response to control computing and storage costs [9]. Modern MeV mission studies, including AMEGO and COSI, similarly rely on Geant4/MEGAlib-based simulations to calculate response, background, event reconstruction, and sensitivity [10–14]. The same philosophy is adopted here, but applied to a narrower and more demanding target: a balloon-borne, focused, narrow-line 511 keV telescope with a TES focal plane.

This paper presents a detector-coupled simulation and sensitivity study for such a payload. The simulation chain starts from a Ge(111) Laue-lens source model at 511 keV and transports focused photons through the detector entrance window into an engineering detector/cryostat mass model. The modeled assembly includes the TES focal plane, staged thermal windows, a Be entrance window, Cu/W passive shielding, cryogenic service proxies, and an active CsI shield. The background chain includes prompt atmospheric components, activation buildup, exact-position delayed-source transport, active-shield vetoes, Compton/field-of-view consistency cuts, a narrow 511 keV line window, mission-time rate evolution, and counting-significance estimates. The main contribution is not an isolated effective-area estimate, but a closed rate calculation in which signal and background pass through the same detector-coupled selection definitions.

The remainder of the paper is organized as follows. Section 2 summarizes the science-driven instrument requirements. Section 3 describes the simulation chain. Sections 4–7 describe the detector geometry, optics bridge, background model, and event selections. Section 8 presents the sensitivity, convergence, and background-composition results. Section 9 discusses interpretation, limitations, and future work. Section 10 summarizes the main conclusions.

107 2. Science and instrument requirements

108 The instrument considered here is optimized for a pointed, narrow-line
109 measurement rather than all-sky imaging. The primary science case is the de-
110 tection or exclusion of compact 511 keV emission at fluxes of order $10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$
111 or below during a long-duration balloon exposure. This flux scale is moti-
112 vated by the need to test bright central-core, compact-source, and transient
113 pair-plasma scenarios while remaining complementary to wide-field Compton
114 surveys.

115 The requirements are stated at the level needed for the present simula-
116 tion, not as final flight-system specifications. They define what the model
117 must preserve for a credible rate claim: the source flux normalization, the fo-
118 cused entrance phase space, the narrow line-window response, and the same
119 detector selection for signal and background. This distinction is important
120 because several quantities, including the TES energy response and final cryo-
121 stat layout, remain design assumptions or engineering proxies.

122 Four requirements follow. First, the telescope must concentrate 511 keV
123 photons so that a small focal plane can be used without sacrificing collect-
124 ing area. Second, the focal-plane response must support a sub-keV analysis
125 window so that the line-window background scales with a narrow spectral
126 interval rather than a broad MeV-band continuum. Third, the detector and
127 shield model must suppress atmospheric, activation, and side-entry back-
128 grounds while explicitly accounting for passive material near the TES ab-
129 sorbers. Fourth, the simulation chain must treat prompt background, delayed
130 activation, focused signal, vetoes, event topology, and mission-time variation
131 consistently. Table 1 summarizes how these requirements map onto the cur-
132 rent v3p5 exact-position implementation.

133 These requirements also set the interpretation of the results. A hard-
134 window sensitivity is the primary claim because it depends only on energy
135 selection, vetoing, topology/FoV consistency, and the mission-time rate fold.
136 Focused-spot and annular sidecars test spatial-analysis boundaries, but they
137 do not replace the hard-window requirement until a publication-grade spatial-
138 spectral likelihood and nuisance-background treatment are in place.

139 3. Simulation workflow

140 Figure 1 summarizes the workflow. The calculation is divided into geom-
141 etry definition, prompt and buildup transport, delayed-source construction,

Table 1: Traceability from science drivers to instrument and simulation requirements. The table summarizes requirements used to define the current v3p5 exact-position analysis; it is not a separate sensitivity calculation.

Science driver	Requirement	Current implementation or evidence	Authority
Compact 511 keV targets at $\sim 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$ or below	Concentrate 511 keV photons onto a small focal plane	Ge(111) f10m A1 Laue lens with $A_{\text{eff}}(511 \text{ keV}) = 20.08476 \text{ cm}^2$; 37,194 within-Be focal rows transported through the detector bridge	Step04, Step09
Narrow-line spectroscopy rather than broad-band counting	Use a sub-keV analysis window and preserve source-flux normalization	$W2 = 510.58\text{--}511.42 \text{ keV}$; Step05 signal $0.00118117 \text{ s}^{-1}$ at $F_{\text{ref}} = 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$; Step07 response $11.8117 \text{ cps}/(\text{ph cm}^{-2} \text{ s}^{-1})$	Step05, Step07
Balloon prompt, side-entry, and activation backgrounds	Apply detector-coupled veto and topology cuts to signal and background streams	CsI post-processing veto, side-entry Compton/FoV consistency, exact-position delayed transport; selected $W2$ background 0.0624651 s^{-1} split into prompt 0.0590827 s^{-1} and delayed $0.00338234 \text{ s}^{-1}$	Step05 exactpos
Mission-level sensitivity claim	Fold rates through mission time and test delayed-source sampling stability	Step08 gives $Z_{20\text{d}} = 6.15522$ and $F_{3\sigma}(20\text{d}) = 4.87391 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$; M/seed sweep changes $Z_{20\text{d}}$ by 0.551%	Step06–08; M/seed

142 focused-optics transport, detector-coupled signal replay, detector event se-
143 lection, mission-time rate folding, source-case normalization, and cumulative
144 significance evaluation. The present manuscript uses the v3p5 full-stat exact-
145 position branch as the primary rate authority. The conservative radial-profile
146 delayed-source branch is retained only as a cross-check.

147 The rate-level authority is the exact-position delayed-source chain labeled
148 `fullstat_v2_exactpos`. It uses full-stat prompt and buildup transport, a
149 fixed day-15 delayed inventory, sampled exact-position `PointSource` delayed
150 support, and full downstream propagation through the Step05–Step08 de-
151 tector response and significance chain. The older homogeneous-beam and
152 legacy delayed-source branches are not used for the headline sensitivities.

153 Table 2 gives the stepwise version of this chain. In the repository, `Project_`
154 `Memory.md` is the audit map that records which files are current numerical

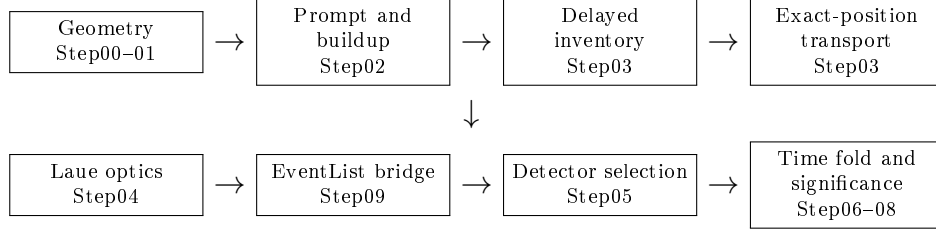


Figure 1: Simulation chain used in this work. The important feature is that the focused signal and all background streams are evaluated in the same detector-response and event-selection space before mission-time significance is calculated.

155 authority, which paths are legacy provenance, and which sidecars are bound-
 156 ary tests. In this manuscript, that internal map is compressed into a re-
 157 producible method description: each step has a defined input, a physical
 158 or statistical operation, an output consumed by later steps, and an explicit
 159 boundary on what may be claimed from it. The B-FULL optics calculation
 160 recorded in the Step04 authority files is one step in this chain, not the chain
 161 itself. Its raw runs come from the `opticsim_full` environment, while the
 162 manuscript-facing authority is the repo-copied Step04 summary, line con-
 163 figuration, rocking-curve map, and focal-crossing table. This optics step
 164 supplies the Laue-lens effective area and tracked focused phase space; the
 165 Step09 EventList bridge and the Step05–Step08 detector-response chain are
 166 what convert that optical phase space into a detector-coupled point-source
 167 sensitivity.

168 Read as an audit framework, the steps have different authority levels.
 169 Steps 00–01 freeze the detector mass model and coordinate policy but do not
 170 produce a rate. Step02 generates prompt and activation-production histories
 171 and defines the exposure normalization. Step03 converts those production
 172 histories into a fixed day-15 delayed inventory and transports sampled exact-
 173 position decays. Step04 is optical only: it computes the focused 511 keV
 174 phase space and effective area but contains no detector background. Step09
 175 is the handoff between optics and detector transport. Step05 is the first
 176 point at which prompt background, delayed background, and focused signal
 177 are compared under the same veto, topology, and line-window rules. Steps
 178 06–08 then move from selected rates to mission-time significance. Step10 is
 179 not a new physics stage; it is the robustness sidecar that tests exact-position
 180 support size and seed dependence at the final selected-rate level.

181 The numerical hierarchy used below follows the same audit logic. Day-15

selected rates are quoted from the Step05 physical-reference-flux block after the unit EventList signal stream has been scaled by the f10m A1 effective area and the reference atmospheric transmission. Mission-mean rates are quoted from Step06, and the headline Z , $T_{3\sigma}$, and flux thresholds are quoted from Step08. Exact-position support-size claims are quoted only after the sampled source has been transported through Step05–Step08. This separation is used to avoid mixing unit-injection bookkeeping rates, day-15 direct expectations, and mission-folded sensitivities in a single headline.

4. Detector and cryostat mass model

The detector model is an engineering Geant4/MEGAlib proxy for a balloon-borne TES focal-plane assembly. It is not intended to be a final CAD reproduction; rather, it captures the material and geometrical features most relevant to MeV background formation and event selection. The current v3p5 center-finger geometry contains a TES pixel array, a Be entrance window, staged Al thermal windows, Cu/W passive shields and collimation, cryogenic service-mass proxies, and a segmented CsI active shield. The local side-entry geometry points the window 45° upward in the zenith-frame source coordinates, with an instrument-frame rotation of 0, 45, 0 degrees. A 2 cm-thick W square-bore side collimator defines the local side-window entrance.

The detector core is represented by six layers of 376 TES pixel copies (2256 copies in total). The side-entry collimator is therefore an engineering aperture proxy rather than a pixel-matched 376-hole collimator. The geometry closure fixes these material and coordinate choices before rate calculations are made; it does not by itself define a sensitivity.

The active shield is treated quantitatively by post-processing rather than by relying on native production-trigger filtering. Candidate events are grouped on a common Poisson time axis and vetoed if the summed CsI shield energy in the coincidence window exceeds the analysis threshold. For the current CsI branch the active-veto threshold is 50 keV and the coincidence window is $1\mu\text{s}$. This choice avoids filtering activation-production storage at the transport stage and keeps the veto definition common to prompt, delayed, and focused-science streams.

5. Laue optics and detector-coupled signal replay

The signal model uses a Ge(111) Laue-lens configuration tuned to 511 keV. The current f10m A1 optical authority has focal length 10 m, 25 tiles, 18

217 mm tile size, 30 arcsec mosaicity, active Ge mass 0.44063 kg, and $A_{\text{eff}}(511 \text{ keV}) =$
 218 20.08476 cm^2 . A three-seed B-FULL focal-crossing aggregate contains 37,194
 219 within-Be focal rows, with within-Be fraction 0.998336, $r_{90} = 1.02853 \text{ cm}$,
 220 and $r_{99} = 1.24527 \text{ cm}$.

221 The optics and detector are coupled through an EventList bridge. Focal-
 222 crossing rows accepted inside the Be-window gate are transformed into the lo-
 223 cal detector coordinate system and replayed through the full detector/cryostat
 224 geometry. This avoids using an analytic phase-space projection as the signal
 225 authority and preserves the optical focusing handoff at the detector entrance
 226 plane. The v3p5 bridge transported 37,194/37,194 EventList rows. The
 227 bridge is a signal handoff only: it does not add cosmic-ray prompt background
 228 or delayed activation in the Laue-lens hardware. At $F_{\text{ref}} = 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$,
 229 the Step05 hard-window signal response is $0.00118117 \text{ s}^{-1}$, corresponding to
 230 a Step07 response of $11.8117 \text{ cps}/(\text{ph cm}^{-2} \text{ s}^{-1})$.

231 6. Prompt and delayed background model

232 The background model contains prompt atmospheric transport and de-
 233 layed activation. The full-stat prompt stream and the activation-buildup
 234 stream each generated 25,210,216 particles. The prompt input is split into
 235 68 SIM/DAT products, and each primary-particle tag is normalized with its
 236 own transported exposure using the rule $r_{\text{event}} = 1/\sum TT_{\text{tag}}$. This per-tag
 237 normalization is important because gamma and non-gamma prompt streams
 238 have different splitting and exposure structure. Delayed activation is con-
 239 structed from the activation-production ledgers and corrected ground-state
 240 half-lives. The fixed day-15 delayed inventory activity for the current chain
 241 is 85.636696 Bq .

242 The main methodological upgrade in the current branch is exact-position
 243 delayed-source transport. Instead of compressing production positions into
 244 axisymmetric radial-profile source blocks, the exact-position chain samples
 245 equal-flux PointSource support blocks from weighted radioactive-production
 246 rows. The baseline exact-position source uses $M = 5000$ support blocks
 247 drawn from 253,770 sampling rows with seed 260613, contains no RadialProfileBeam
 248 blocks, and preserves the fixed activity budget. The delayed transport used
 249 10^6 triggers and stored 10^6 events, with observation time 11530.473845 s .

250 This exact-position source representation is not a one-block-per-production-
 251 row stress test. Instead, it is a statistically sampled representation whose
 252 adequacy is evaluated downstream in Section 8.3. The convergence test is

performed on the final selected line-window rates and significance, not only on source-sampling diagnostics.

7. Event selection and significance calculation

The analysis window is

$$W2 = [510.58, 511.42] \text{ keV}, \quad (1)$$

corresponding to $511 \text{ keV} \pm 420 \text{ eV}$. This window is matched to a sub-keV TES response assumption and is narrow enough to suppress continuum background while retaining most of the line response.

Events are processed on a common time axis. Prompt, delayed, and focused signal event instances are drawn according to their rates, sorted by time, and grouped within a $1 \mu\text{s}$ coincidence window. The active-shield veto is then applied to the summed shield energy. TES hit topologies are classified with a custom Compton/field-of-view consistency test. One-hit events are retained. Two-hit events test both possible first-scatter orders. Higher-multiplicity events enumerate candidate orders and use a Compton sequence quality metric before testing whether the first-scatter cone is consistent with the Be-window acceptance. Events without a valid reconstruction are retained under the current conservative rejection policy, so the quoted hard-window sensitivity does not rely on an aggressive reconstruction failure veto.

The time-dependent significance is computed as a cumulative counting statistic. For a reference flux F_{ref} , the cumulative significance is

$$Z(t) = \frac{N_s(t)}{\sqrt{N_b(t)}}, \quad (2)$$

where $N_s(t)$ and $N_b(t)$ are the time-integrated selected signal and background counts after mission-time scaling and live-factor corrections. For comparison among analysis variants, the 3σ flux threshold is scaled as

$$F_{3\sigma}(t) = F_{\text{ref}} \frac{3}{Z(t)}. \quad (3)$$

This Gaussian counting convention is used consistently for the hard-window results and sidecar comparisons. The Step05 summary also records a day-15 direct expectation. The manuscript headline does not use that constant-rate direct value; it uses the Step06 mission-time fold and the Step08 cumulative significance. The distinction is small for the present hard-window case, but keeping it explicit is necessary for comparing branches and sidecars.

282 8. Results

283 8.1. Primary hard-window sensitivity

284 Table 4 gives the primary hard-window result. In the exact-position chain,
 285 the Step05 selected $W2$ background and signal rates at $F_{\text{ref}} = 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$
 286 are 0.0624651 s^{-1} and $0.00118117 \text{ s}^{-1}$, respectively. After the mission-time
 287 fold, the mean background and signal rates are 0.0626835 s^{-1} and $0.00117281 \text{ s}^{-1}$.
 288 The resulting 20-d significance is $Z_{20\text{d}} = 6.15522$, with a 3σ detection time
 289 of 4.73758 d. The 20-d 3σ flux threshold is $4.87391 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$, and the
 290 1 Ms threshold is $6.32564 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$.

291 The hard-window result is intentionally conservative relative to the spa-
 292 tial sidecars. It requires only the narrow energy window, active-shield veto,
 293 and topology/FoV consistency selection. The standalone r_{90} sidecar uses
 294 the radial-profile full-stat branch and illustrates the possible gain from us-
 295 ing the focused spot in the local side-window detector frame. The exact-
 296 position boundary sidecars apply a 45° line-of-sight atmosphere fold and a
 297 fixed-template annular spatial likelihood. These rows should be treated as
 298 boundary tests and post-processing analyses, not as finalized nuisance-profile
 299 likelihood results.

300 8.2. Background composition

301 The selected exact-position $W2$ background is dominated by prompt at-
 302 mospheric events. The Step05 stream split is 0.0590827 s^{-1} prompt and
 303 $0.00338234 \text{ s}^{-1}$ delayed, corresponding to 94.6% and 5.4% of the total selected
 304 hard-window background. The largest prompt component is the positron-
 305 tagged atmospheric stream, contributing 0.0543377 s^{-1} or 86.99% of the total
 306 $W2$ background. The delayed component is led by ^{64}Cu , which contributes
 307 $0.00251507 \text{ s}^{-1}$, or 74.36% of the delayed $W2$ events.

308 This composition has two implications. First, improving the hard-window
 309 sensitivity is primarily a prompt-background problem for the current geom-
 310 etry and selection. Second, delayed activation is small but not negligible;
 311 its line-window contribution is sufficiently structured that exact-position de-
 312 layed transport is worth preserving in the chain.

313 8.3. Exact-position support-size and seed convergence

314 Table 6 summarizes the transport-backed exact-position convergence sweep.
 315 Four downstream cases were run: two seeds at $M = 5000$ and two larger sup-
 316 port sizes at $M = 20000$ and $M = 50000$. Across these cases, the delayed

subcomponent shows visible Monte Carlo variation, but the total selected $W2$ background changes by only 1.119%, and Z_{20d} changes by only 0.551%. The final sensitivity is therefore stable at the level required for the present rate claim.

8.4. *Equal-attenuation BGO active-shield control*

An active-shield material control was run with a BGO shield adjusted to approximately match the CsI attenuation budget in the current geometry. The BGO sample uses side, bottom, and top active thicknesses of 2.137, 3.287, and 1.582 cm, respectively, an adaptive outer Al shell around the new active envelope, and a 70 keV BGO active-veto threshold. The equal-attenuation check has a maximum absolute relative difference of 0.0731, and the BGO active mass is 45.0259 kg, compared with 62.8337 kg for the source CsI active shield.

The BGO branch was propagated through the same hard-window Step05–Step08 rate chain as the CsI exact-position authority. It includes full-stat prompt and buildup transport, a 5000-block exact-position delayed source drawn from true RPIP rows, delayed transport with 1,000,000 stored events, and detector replay of the f10m A1 focused EventList. Table 7 summarizes the matched comparison. Relative to the CsI exact-position chain, the BGO sample lowers the $W2$ mission-mean background by 7.738%, raises the 20-d significance by 4.541%, and lowers the 20-d 3σ flux threshold by 4.344%. This is a hard-window material-control result only; no BGO spatial/profile-likelihood sidecar or threshold/material-uncertainty scan is included.

8.5. *Comparison with instrument benchmarks*

The present calculation is not an all-sky imaging forecast. It is a pointed, narrow-line, compact-source sensitivity estimate. The comparison most relevant to its design philosophy is therefore not simply field-of-view survey speed, but line-window background suppression per focused point-source flux. INTEGRAL/SPI and COSI remain the key references for mapping the diffuse 511 keV sky [2–5]. COSI’s satellite implementation will survey the 0.2–5 MeV sky with wide field of view and Ge detector spectroscopy; NASA and HEASARC mission pages list a planned 2027 launch and a 6.0 keV energy resolution at 511 keV [15, 16]. The TES Laue-lens concept considered here is complementary: it trades sky coverage for a concentrated beam and a sub-keV line window.

Relative to a wide-field Compton telescope, the proposed payload would be most competitive for compact 511 keV targets, central-core searches, and transient follow-up observations where the source position is known or constrained. It is less suited to measuring the full diffuse bulge and disk luminosity, because the small focused aperture samples only a small solid angle. This distinction is essential when comparing point-source Route-A results with diffuse-sky Route-B foreground estimates.

9. Discussion

The main result is that a focused TES line spectrometer can reach a 20-d 3σ point-source threshold below $5 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$ in the current hard-window exact-position chain. This threshold is obtained without invoking the focused-spot spatial sidecar as the headline result. It follows from three combined effects: optical concentration, a sub-keV energy window, and a detector-coupled veto/topology selection that suppresses prompt and delayed backgrounds in the same selection space as the signal.

The exact-position delayed-source treatment is a methodological improvement over radial-profile compression. Although delayed activation contributes only about 5.4% of the selected hard-window background in the present chain, it is spatially and materially structured. Preserving sampled production positions prevents artificial azimuthal or radial redistribution from becoming an unquantified systematic. The downstream convergence sweep indicates that $M = 5000$ support blocks are sufficient for the current hard-window rate claim, while larger support-size stress tests remain useful engineering checks.

The background composition points to the next optimization targets. The selected hard-window rate is dominated by prompt atmospheric e^+ -tagged events. Improvements may therefore come from shield geometry, side-entry collimation, material reduction near the focal plane, stricter topology cuts, or improved reconstruction of multi-pixel events. Delayed activation is smaller but led by identifiable nuclides and volumes, suggesting targeted material substitutions or local shielding may still be valuable.

Several limitations should be kept explicit. First, the detector/cryostat model is an engineering proxy, not a final flight CAD model. Second, the production SIM files use a post-processing active-shield authority; native trigger/veto production filtering is not the source of quantitative rates. Third, the Step06 mission-time calculation is a rate-level fold, not a new particle

transport at each trajectory time bin. Fourth, the custom Compton/FoV veto is not yet a full Revan/Mimrec reconstruction. Fifth, upstream Laue-lens self-background and activation from cosmic-ray interactions in the optics hardware are not included in the headline detector-background budget. Sixth, spatial sidecars such as r_{90} and multi-annulus likelihood estimates are promising, but they are not yet a publication-grade nuisance-profile likelihood. Seventh, the BGO control in Section 8.4 is a hard-window material comparison under its 70 keV shield threshold. It does not include BGO spatial/profile-likelihood sidecars or threshold/material-uncertainty scans, so the CsI exact-position chain remains the manuscript headline.

These limitations do not invalidate the primary conclusion, but they define the next work required before a final instrument proposal or flight performance paper. The most valuable follow-up studies are a Revan cross-check of the multi-hit subset, an optics self-background simulation, a final cryostat/DR geometry update, measured or literature-anchored TES saturation and pile-up studies at the relevant rates, and a spatial-spectral likelihood that profiles background nuisance parameters.

10. Conclusions

We have presented a first full-chain detector-coupled background and sensitivity study for a balloon-borne pointed 511 keV TES Laue-lens telescope. The current v3p5 exact-position chain couples a Ge(111) Laue optical response to a TES focal-plane mass model, prompt atmospheric transport, delayed activation, active-shield vetoing, Compton/FoV topology selection, mission-time rate folding, and counting-significance evaluation. In the primary 510.58–511.42 keV hard window, the chain predicts $B = 0.0624651 \text{ s}^{-1}$ and $S = 0.00118117 \text{ s}^{-1}$ at $10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$, giving $Z_{20d} = 6.15522$ and $F_{3\sigma}(20d) = 4.87391 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$. Exact-position delayed-source convergence is stable at the final hard-window significance level. The result supports the feasibility of focused TES spectroscopy as a complementary 511 keV technique for compact-source and central-core searches, while identifying prompt atmospheric background suppression and reconstruction validation as the most important next steps.

CRediT authorship contribution statement

Placeholder: to be completed with the final author list.

422 Declaration of competing interest

423 The authors declare no competing financial interests. This statement
424 should be verified by all authors before submission.

425 Data and software availability

426 Simulation cards, validation summaries, and analysis scripts will be made
427 available in a public repository or institutional archive before publication.
428 Internal path names in this draft are placeholders and should be replaced by
429 persistent identifiers.

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Table 2: Stepwise simulation workflow used for the v3p5 exact-position rate authority. The table is a manuscript-facing compression of the repository workflow; it does not replace the validation files that define the numerical authority.

Step	Role in the calculation	Current implementation or evidence	Boundary for interpretation
00–01 Geometry	Define detector mass model, active volumes, local side-entry frame, and injection surfaces	v3p5 center-finger geometry with TES array, Be side window, Cu/W passive material, and segmented CsI active shield; local window tilted 45° upward	Geometry closure only; not a rate calculation
02 Prompt and buildup transport	Generate prompt atmospheric events and activation-production histories	Full-stat v2 prompt and buildup streams, each with 25,210,216 generated particles; prompt rates use per-tag exposure normalization	Low-stat branches are closure checks, not paper-facing rate authority
03 Delayed inventory and source	Convert activation products to a fixed day-15 source and transport delayed decays	Fixed inventory activity 85.636696 Bq; exact-position sampled PointSource support with $M = 5000$ and 10^6 delayed transport triggers	Sampled support is activity preserving; convergence is tested downstream in Step10
04 B-FULL Laue optics	Compute the 511 keV optical effective area and focused focal-plane phase space	Step04 B-FULL Ge(111) authority files, derived from raw opticsim_full runs with external XOP/CRYSTAL rocking curves; f10m Al gives $A_{\text{eff}}(511) = 20.08476 \text{ cm}^2$ and 37,194 within-Be focal rows	Optical signal calculation only; no upstream optics self-background is added
09 EventList bridge	Replay tracked optical focal crossings through the detector geometry	37,194/37,194 Step04 rows are written as a MEGAlib EventList and transported through the v3p5 detector setup	Uses tracked focal crossings, not an analytic phase-space projection
05 Detector response	Apply common detector selection to prompt, delayed, and focused streams	Active-shield post-processing veto, side-entry Compton/FoV consistency, and W2 hard-window selection; day-15 W2 rates $B/S = 0.0624651/0.00118117 \text{ s}^{-1}$	Quantitative veto authority is the post-processing model, not native production filtering
06 Mission-time fold	Propagate selected rates through balloon mission time	Time-dependent prompt, delayed-activity, atmosphere, and accidental-veto live-factor fold; mission-mean W2 rates $B/S = 0.0626835/0.00117281 \text{ s}^{-1}$	Rate-level fold; not a new particle transport at each time bin
07 Source-case ledger	Map detector response to source-flux units	W2 response 11.8117 cps/(ph cm ⁻² s ⁻¹) for the reference point-source case	Keeps Route-A point-source normalization separate from diffuse-sky sidecars
08 Significance	Compute cumulative detection significance and flux thresholds	Time-dependent hard-window result: $Z_{20d} = 6.15522$, $T_{3\sigma} = 4.73758 \text{ d}$, and $F_{3\sigma}(20d) = 4.87391 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$	Gaussian counting metric used for consistent comparisons
10 Robustness sidecar	Test exact-position support size and seed dependence at final selected-rate level	Four transport-backed cases over $M = 5000, 20,000, \text{ and } 50,000$; W2 background and Z_{20d} relative ranges are 1.119% and 0.551%	Supports promoting exact-position delayed transport as the current rate authority

Table 3: Key detector and analysis parameters used for the primary v3p5 exact-position chain.

Parameter	Value or policy
Detector concept	TES microcalorimeter focal plane
TES pixel copies	$6 \times 376 = 2256$
Active shield	Segmented CsI, post-processing veto
CsI veto threshold	50 keV summed shield energy
Entrance window	Be side window
Passive shielding/collimation	Cu/W proxy shields and 2 cm W side sleeve
Pointing	Local side window tilted 45° upward
Detector trigger/veto authority	Step05 post-processing selection
Coincidence window	$1 \mu\text{s}$
Line window W_2	510.58–511.42 keV
Reference flux F_{ref}	$10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$

Table 4: Primary sensitivity and selected comparison cases. The exact-position hard-window row is the current rate authority. The radial-profile row is a conservative baseline cross-check. Sidecar rows state their base branch explicitly and are not replacements for the hard-window authority.

Case	Base	B_{W_2} (cps)	S_{W_2} at F_{ref} (cps)	$Z_{20\text{d}}$	$T_{3\sigma}$ (d)	$F_{3\sigma}(20\text{d})$
Exact-position hard window	exactpos	0.0624651	0.00118117	6.15522	4.73758	4.87391×10^{-5}
Radial-profile hard window	radial	0.0729576	0.00118117	5.70221	5.46687	5.26111×10^{-5}
Focused r_{90} spatial sidecar	radial	0.0232510	≈ 0.001063	8.17566	2.23643	3.66943×10^{-5}
45° LOS hard-window sidecar	exactpos	–	–	5.42468	5.93004	5.53028×10^{-5}
45° LOS r_{90} sidecar	exactpos	–	–	7.20513	2.85677	4.16370×10^{-5}
Fixed-template annular sidecar	exactpos	–	–	8.45781	–	3.54702×10^{-5}

Dash entries indicate sidecar quantities that are not independent Step05 day-15 transported rates. The 45° LOS rows are derived by applying the mission-time atmospheric slant-column model to the exact-position chain; the annular row is a fixed-template post-processing likelihood evaluated with selected W2 event rates and is reported only by its final significance and flux threshold.

Table 5: Selected $W2$ background composition in the exact-position chain. Percentages are relative to the total $W2$ background unless otherwise stated.

Component	Rate (cps)	Fraction
All prompt atmospheric streams	0.0590827	94.6%
All delayed activation streams	0.00338234	5.4%
Prompt e^+ stream	0.0543377	86.99%
Delayed ^{64}Cu	0.00251507	74.36% of delayed

Table 6: Transport-backed exact-position convergence cases.

Label	M	Seed	B_{W2} (cps)	Z_{20d}	$F_{3\sigma}(20d)$
fullstat_v2_exactpos	5000	260613	0.0624651	6.15522	4.87391×10^{-5}
m05000_s260614	5000	260614	0.0631683	6.12141	4.90083×10^{-5}
m20000_s260613	20000	260613	0.0627261	6.14261	4.88392×10^{-5}
m50000_s260613	50000	260613	0.0629804	6.13039	4.89365×10^{-5}

Table 7: Matched CsI and equal-attenuation BGO hard-window comparison.

Metric	CsI	BGO	Change
Mission-mean B_{W2} (cps)	0.0626835	0.0578330	-7.738%
Mission-mean S_{W2} at F_{ref} (cps)	0.00117281	0.00117756	+0.405%
Step08 Z_{20d}	6.15522	6.43475	+4.541%
Step08 $T_{3\sigma}$ (d)	4.73758	4.21622	-11.005%
Step08 $F_{3\sigma}(20d)$	4.87391×10^{-5}	4.66219×10^{-5}	-4.344%